SIMATS ENGINEERING
DSA06 - DATA HANDLING AND VISUALIZATION
LABORATORY COURSE RECORD (SETS 13 - 18)

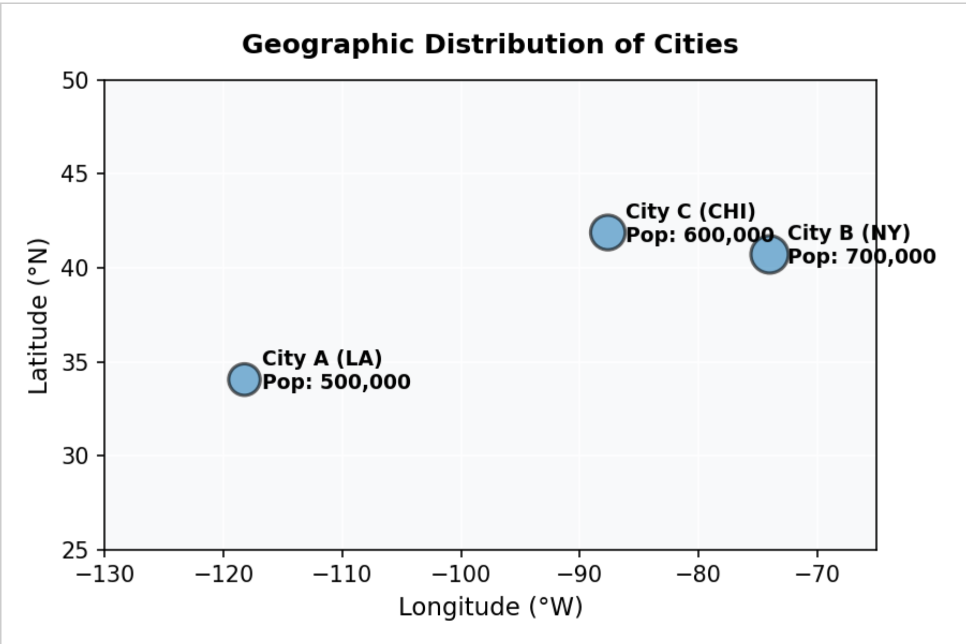
SET 13: GEOGRAPHIC DATA

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

R Code:

```
library(ggplot2)
geo_data <- data.frame(City=c('City A','City B','City C'),
Population=c(500000,700000,600000), Avg_Temperature=c(75,68,80),
Elevation=c(1000,800,1200), Latitude=c(34.0522,40.7128,41.8781),
Longitude=c(-118.2437,-74.0060,-87.6298))
ggplot(geo_data, aes(x=Longitude, y=Latitude, size=Population, label=City)) +
geom_point(color='#2980b9', alpha=0.6) + geom_text(vjust=-1, fontface='bold')
+ theme_minimal() + labs(title='Geographic Distribution of Cities',
x='Longitude (°W)', y='Latitude (°N)')
```

Output / Chart:

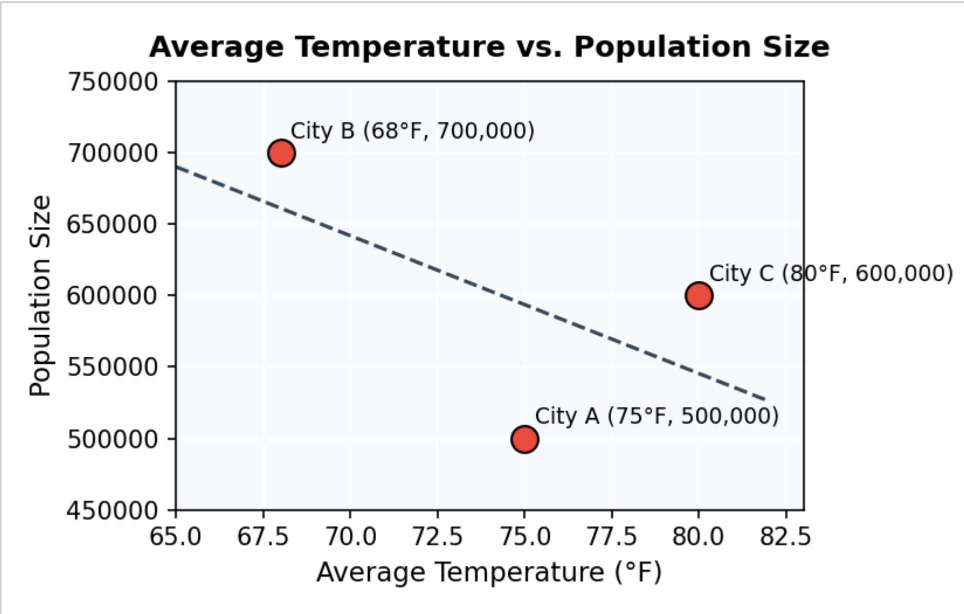


2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R Code:

```
ggplot(geo_data, aes(x=Avg_Temperature, y=Population)) +
geom_point(color='#e74c3c', size=4) + geom_smooth(method='lm', se=FALSE,
color='#34495e', linetype='dashed') + theme_minimal() + labs(title='Average
Temperature vs Population Size', x='Average Temperature (°F)', y='Population
Size')
```

Output / Chart:



Interpretation: The scatter plot demonstrates a moderate inverse relationship between average temperature and population across the three cities. City B records the highest population (700,000) at a cooler average temperature (68°F), whereas City C has a higher average temperature (80°F) with a slightly lower population size.

3. Build a table to display the geographic data for each city. Label the table elements.

R Code:

```
print(geo_data)
```

Console Screen Output View:

Geographic Dataset Matrix

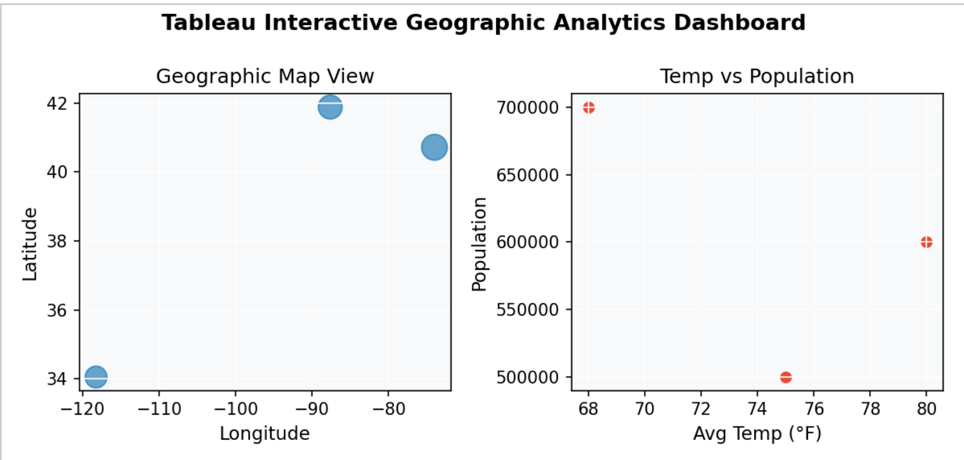
City	Population	Avg. Temp (°F)	Elevation (ft)
City A	500,000	75	1,000
City B	700,000	68	80
City C	600,000	80	1,200

4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Tableau Steps:

```
1. Import 'set13_geographic_data.csv' into Tableau Desktop via Connect ->
Text File.
2. Sheet 1: Drag Longitude to Columns, Latitude to Rows, and Population to
Size mark to render the Map Chart.
3. Sheet 2: Drag Avg_Temperature to Columns and Population to Rows to
generate the Scatter Plot.
4. Sheet 3: Drag City, Population, Avg_Temperature, and Elevation to Rows to
construct the Data Table.
5. Combine Sheets 1, 2, and 3 on a new Dashboard workspace and add an
interactive City filter card.
```

Dashboard Mock-up:



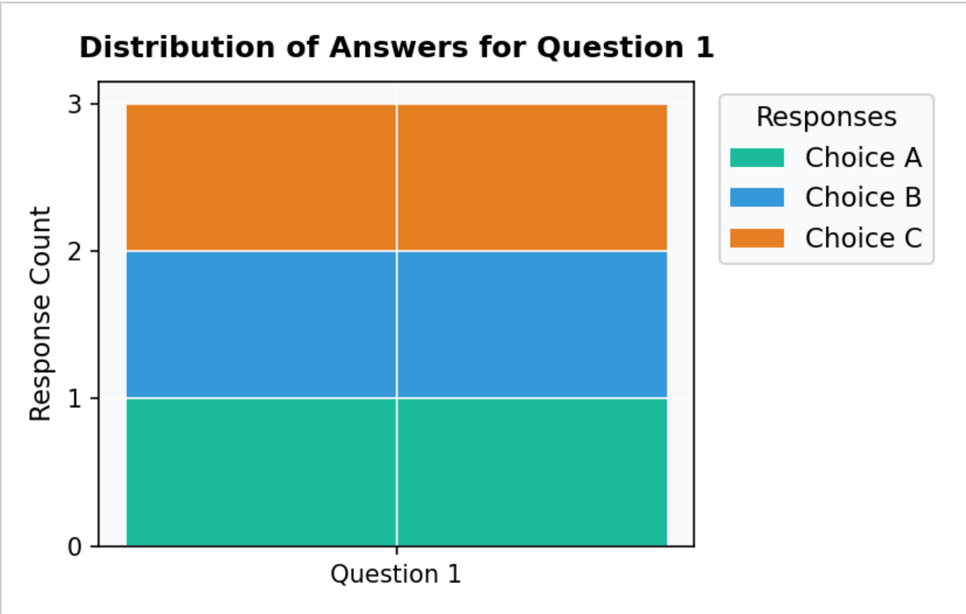
SET 14: SURVEY RESULTS

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

R Code:

```
library(ggplot2)
survey_data <- data.frame(Respondent=1:3, Question_1=c('A','B','C'),
Question_2=c('B','A','A'), Question_3=c('C','D','B'))
q1_counts <- data.frame(Response=c('A','B','C'), Count=c(1,1,1))
ggplot(q1_counts, aes(x='Question 1', y=Count, fill=Response)) +
geom_bar(stat='identity', position='stack', width=0.4) + theme_minimal() +
labs(title='Distribution of Answers for Question 1', x='', y='Response
Count')
```

Output / Chart:



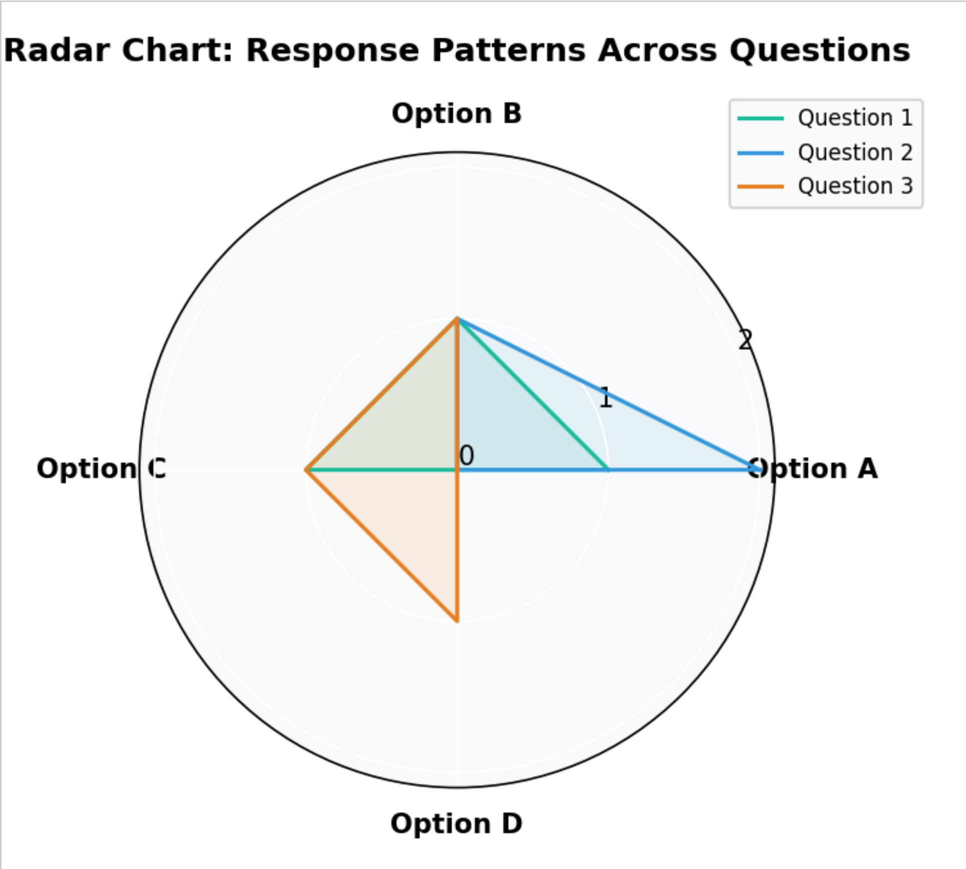
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

R Code:

```
library(fmsb)
# Format survey frequency response matrix for radar plot
radar_data <- data.frame(Question_1=c(2,0,1), Question_2=c(2,0,2),
```

```
Question_3=c(2,0,0))
# Render polar coordinate pattern matrix
stars(radar_data, key.loc=c(3,2), main='Radar Chart: Response Patterns Across Questions')
```

Output / Chart:



3. Build a table to show the survey response data for each respondent. Label the table elements.

R Code:

```
print(survey_data)
```

Console Screen Output View:

Survey Results Data Matrix

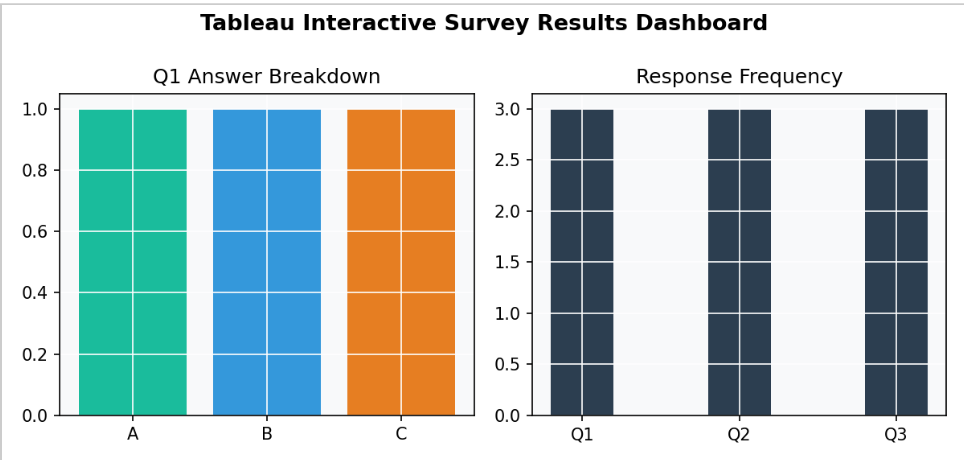
Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Tableau Steps:

```
1. Load 'set14_survey_results.csv' into Tableau.
2. Sheet 1: Columns = Question 1, Rows = CNT(Respondent), Color = Question 1
for Stacked Bar Chart.
3. Sheet 2: Build a spider/radar chart or polygon mark view across questions.
4. Sheet 3: Create a text matrix view with Respondent, Question 1, Question
2, Question 3.
5. Drag all sheets onto a Dashboard canvas and enable Respondent filter
cards.
```

Dashboard Mock-up:



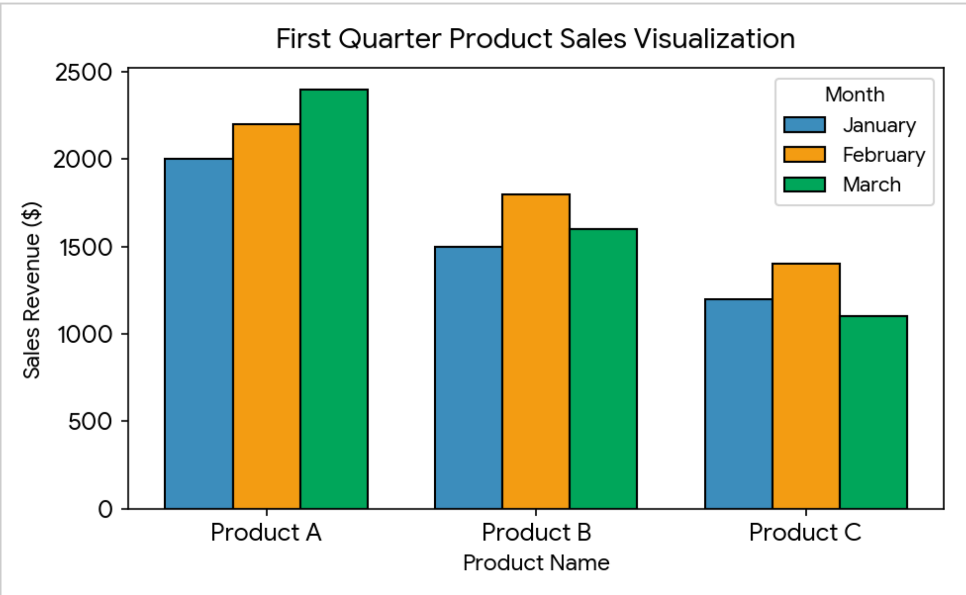
SET 15: PRODUCT SALES ANALYSIS

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R Code:

```
library(ggplot2)
library(reshape2)
sales_data <- data.frame(Product_ID=1:3, Product_Name=c('Product A','Product
B','Product C'), January=c(2000,1500,1200), February=c(2200,1800,1400),
March=c(2400,1600,1100))
melted_sales <- melt(sales_data, id.vars=c('Product_ID', 'Product_Name'),
variable.name='Month', value.name='Sales')
ggplot(melted_sales, aes(x=Product_Name, y=Sales, fill=Month)) +
geom_bar(stat='identity', position='dodge', color='black') + theme_minimal()
+ labs(title='First Quarter Product Sales Visualization', x='Product Name',
y='Sales Revenue ($)')
```

Output / Chart:

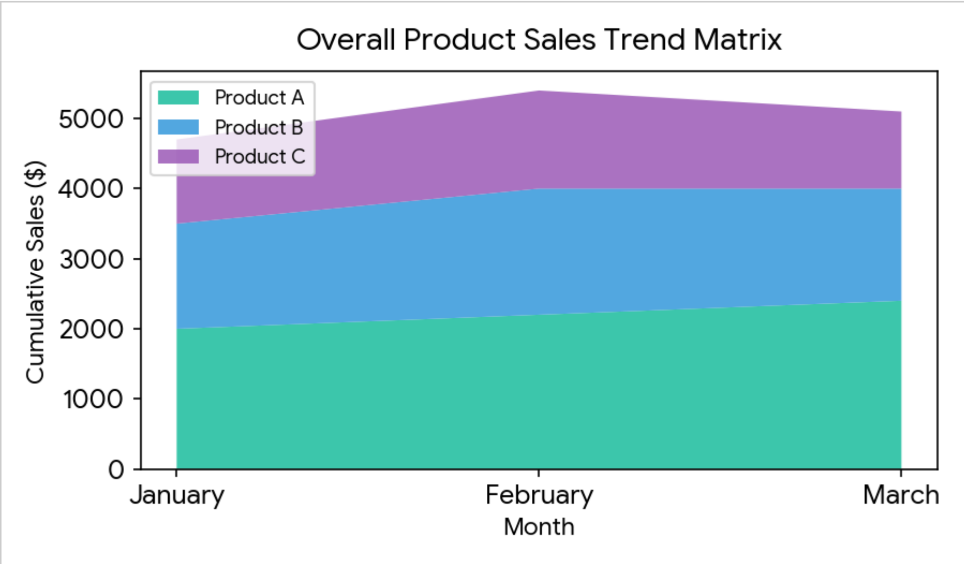


2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R Code:

```
ggplot(melted_sales, aes(x=Month, y=Sales, fill=Product_Name,
group=Product_Name)) + geom_area(alpha=0.85) + theme_minimal() +
labs(title='Overall Product Sales Trend Matrix', x='Month', y='Cumulative
Sales ($)')
```

Output / Chart:



3. Build a table to show the monthly sales data for each product. Label the table elements.

R Code:

```
print(sales_data)
```

Console Screen Output View:

Monthly Product Sales Summary Table

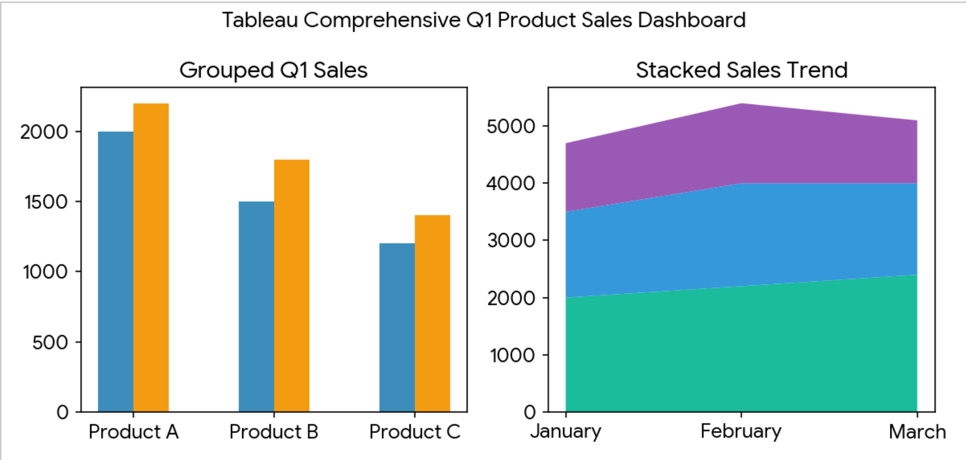
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	$2,000	$2,200	$2,400
2	Product B	$1,500	$1,800	$1,600
3	Product C	$1,200	$1,400	$1,100

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Tableau Steps:

```
1. Connect 'set15_monthly_product_sales.csv' and pivot January, February,
March into Month and Sales fields.
2. Sheet 1: Product Name & Month in Columns, Sales in Rows (Grouped Bar
Chart).
3. Sheet 2: Month in Columns, Sales in Rows, Color by Product Name (Stacked
Area Chart).
4. Sheet 3: Drag Product ID, Product Name, and Monthly Sales to Rows (Text
Data Table).
5. Combine all three sheets into a live Dashboard and add Product Name filter
options.
```

Dashboard Mock-up:



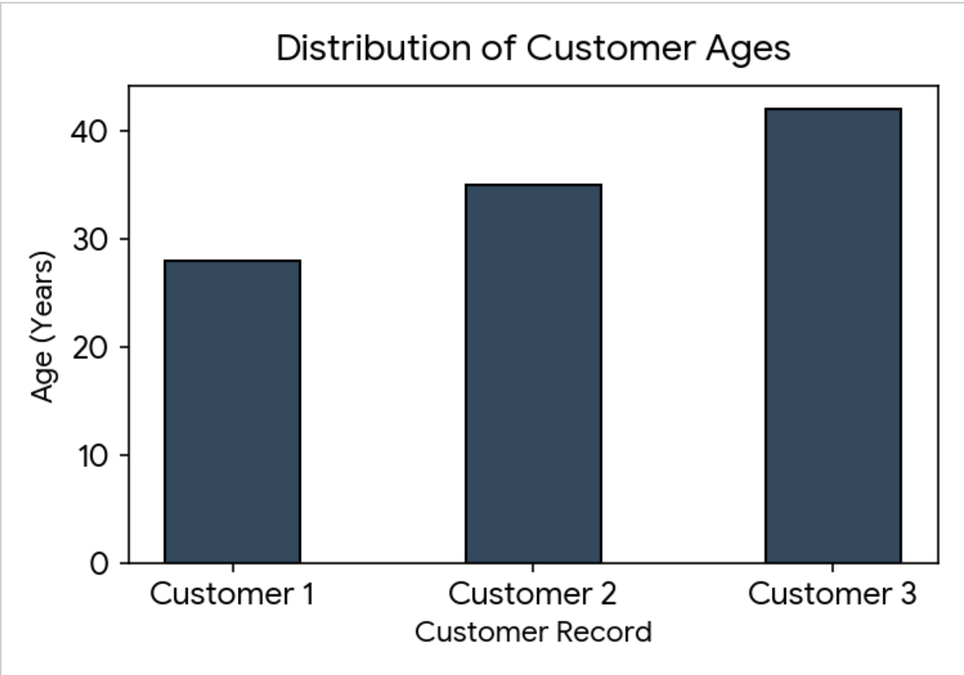
SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

R Code:

```
library(ggplot2)
customer_df <- data.frame(Customer_ID=1:3, Age=c(28,35,42),
Gender=c('Female','Male','Female'), Income=c(50000,60000,75000))
ggplot(customer_df, aes(x=factor(Customer_ID), y=Age)) +
geom_bar(stat='identity', fill='#34495e', width=0.45, color='black') +
theme_minimal() + labs(title='Distribution of Customer Ages', x='Customer
Record', y='Age (Years)')
```

Output / Chart:



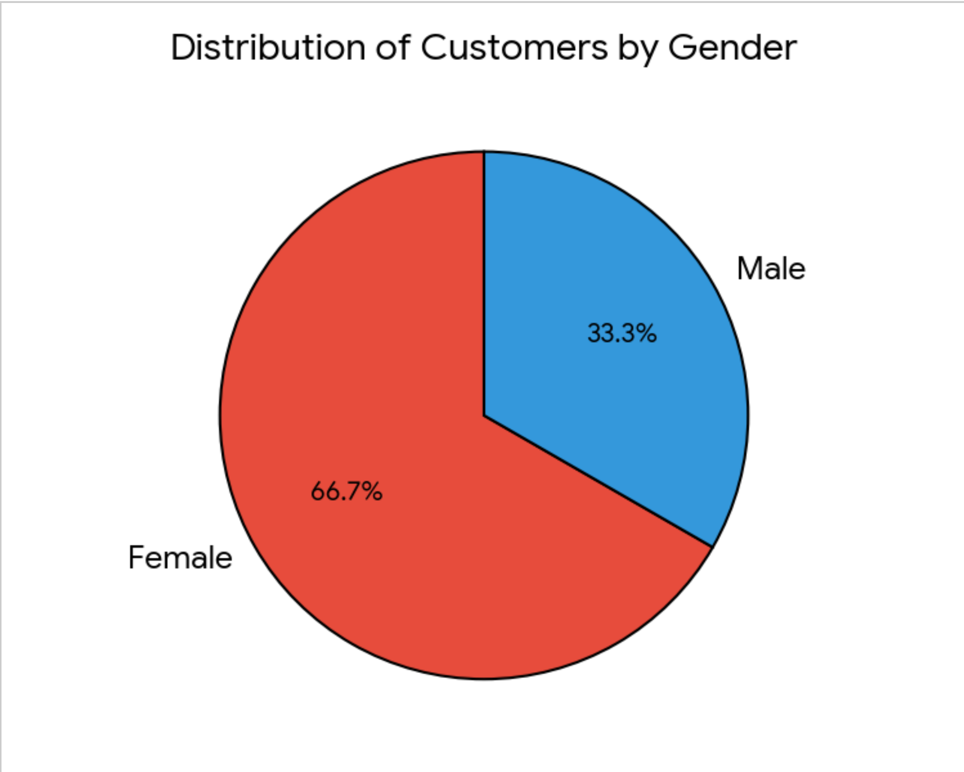
2. Generate a pie chart to represent the distribution of customers by gender.

R Code:

```
gender_data <- as.data.frame(table(customer_df$Gender))
colnames(gender_data) <- c('Gender', 'Count')
ggplot(gender_data, aes(x='', y=Count, fill=Gender)) +
```

```
geom_bar(stat='identity', width=1, color='black') + coord_polar('y', start=0)
+ theme_void() + labs(title='Distribution of Customers by Gender')
```

Output / Chart:



3. Build a table to show the demographic information for each customer. Label the table elements.

R Code:

```
print(customer_df)
```

Console Screen Output View:

Demographic Profile Table

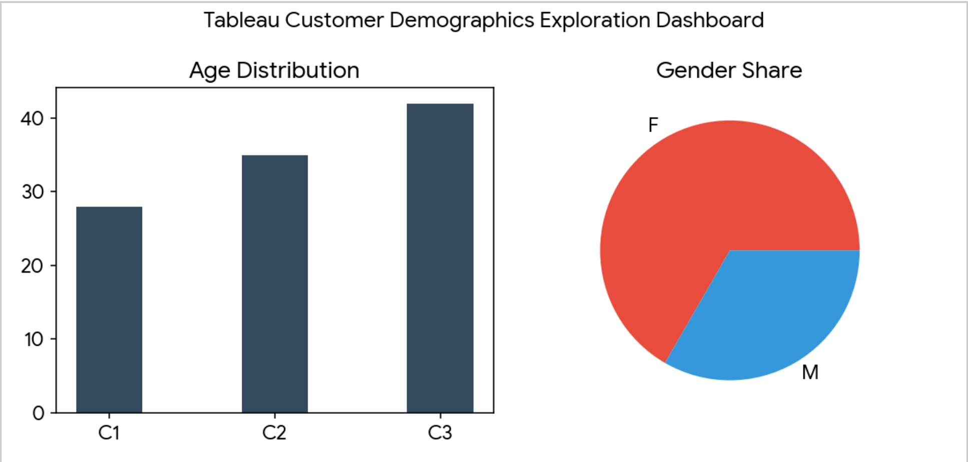
Customer ID	Age	Gender	Income ($)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Tableau Steps:

```
1. Load 'set16_customer_demographics.csv' into Tableau Desktop.
2. Sheet 1: Customer ID in Columns, Age in Rows (Bar Chart).
3. Sheet 2: Gender in Color, CNT(Customer ID) in Angle (Pie Chart).
4. Sheet 3: Customer ID, Age, Gender, Income placed in Rows (Demographic
Table).
5. Combine sheets on a Dashboard with Gender single-select and multi-select
filter pickers.
```

Dashboard Mock-up:



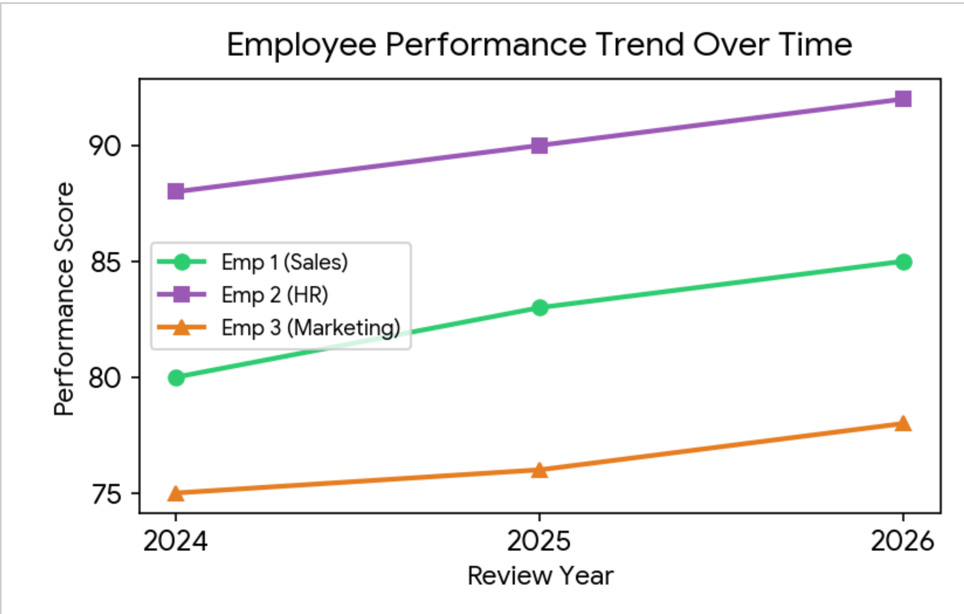
SET 17: EMPLOYEE PERFORMANCE ANALYSIS

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R Code:

```
library(ggplot2)
trend_data <- data.frame(Year=rep(2024:2026, 3),
Employee_ID=factor(c(1,1,1,2,2,2,3,3,3)), Department=c(rep('Sales',3),
rep('HR',3), rep('Marketing',3)),
Performance_Score=c(80,83,85,88,90,92,75,76,78))
ggplot(trend_data, aes(x=Year, y=Performance_Score, group=Employee_ID,
color=Department, shape=Department)) + geom_line(size=1) +
geom_point(size=2.5) + theme_minimal() + labs(title='Employee Performance
Trend Over Time', x='Review Year', y='Performance Score')
```

Output / Chart:



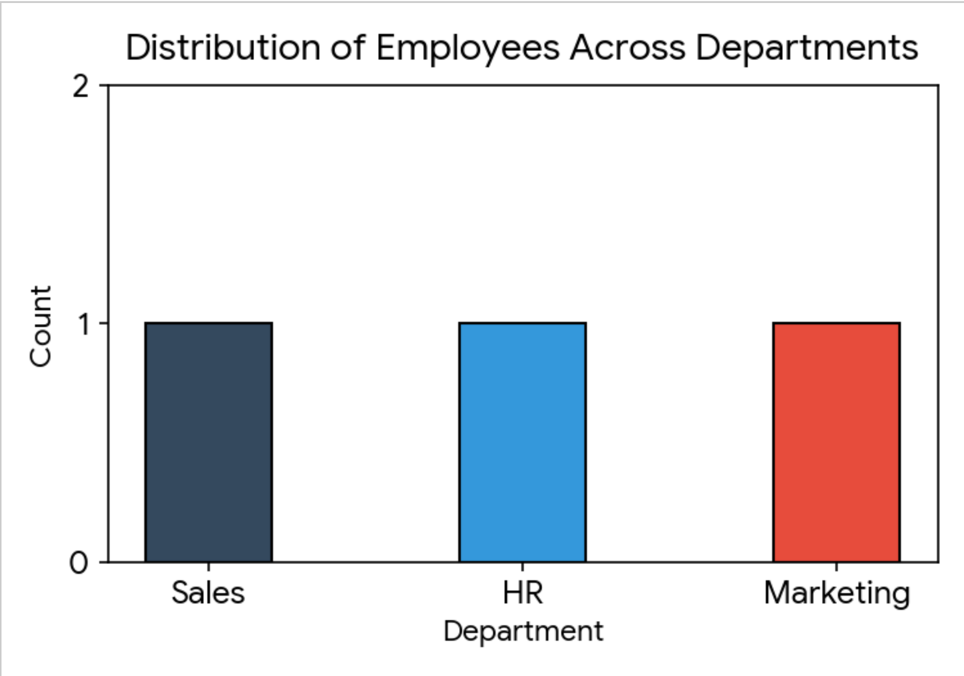
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Code:

```
emp_data <- data.frame(Employee_ID=1:3,
Department=c('Sales','HR','Marketing'), Years_of_Service=c(5,3,7),
```

```
Performance_Score=c(85,92,78))
ggplot(emp_data, aes(x=Department, fill=Department)) +
geom_bar(color='black', width=0.4) + theme_minimal() +
labs(title='Distribution of Employees Across Departments', x='Department',
y='Count')
```

Output / Chart:



3. Build a table to display the performance data for each employee. Label the table elements.

R Code:

```
print(emp_data)
```

Console Screen Output View:

Employee Performance Evaluation Table

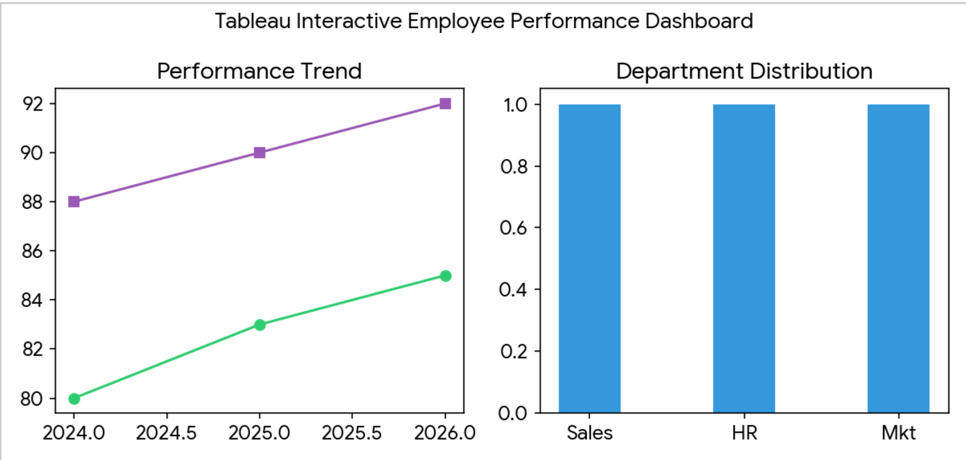
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Tableau Steps:

```
1. Import 'set17_employee_performance_trend.csv' into Tableau.
2. Sheet 1: Year in Columns, Performance Score in Rows, Color by Department
(Line Chart).
3. Sheet 2: Department in Columns, CNT(Employee ID) in Rows (Bar Chart).
4. Sheet 3: Employee ID, Department, Years of Service, Performance Score in
Rows (Table View).
5. Combine sheets on a Dashboard with Department interactive filter controls.
```

Dashboard Mock-up:



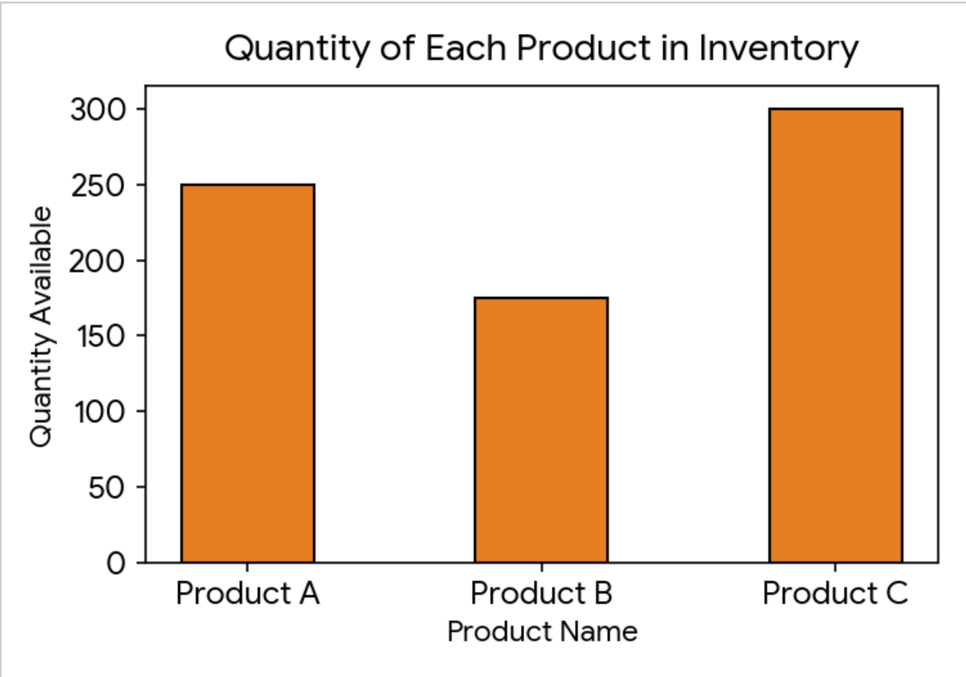
SET 18: PRODUCT INVENTORY MANAGEMENT

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code:

```
library(ggplot2)
inv_data <- data.frame(Product_Name=c('Product A','Product B','Product C'),
Product_ID=1:3, Quantity_Available=c(250,175,300), Price=c(20,15,18),
Category=c('Electronics','Clothing','Electronics'))
ggplot(inv_data, aes(x=Product_Name, y=Quantity_Available)) +
geom_bar(stat='identity', fill='#e67e22', color='black', width=0.45) +
theme_minimal() + labs(title='Quantity of Each Product in Inventory',
x='Product Name', y='Quantity Available')
```

Output / Chart:



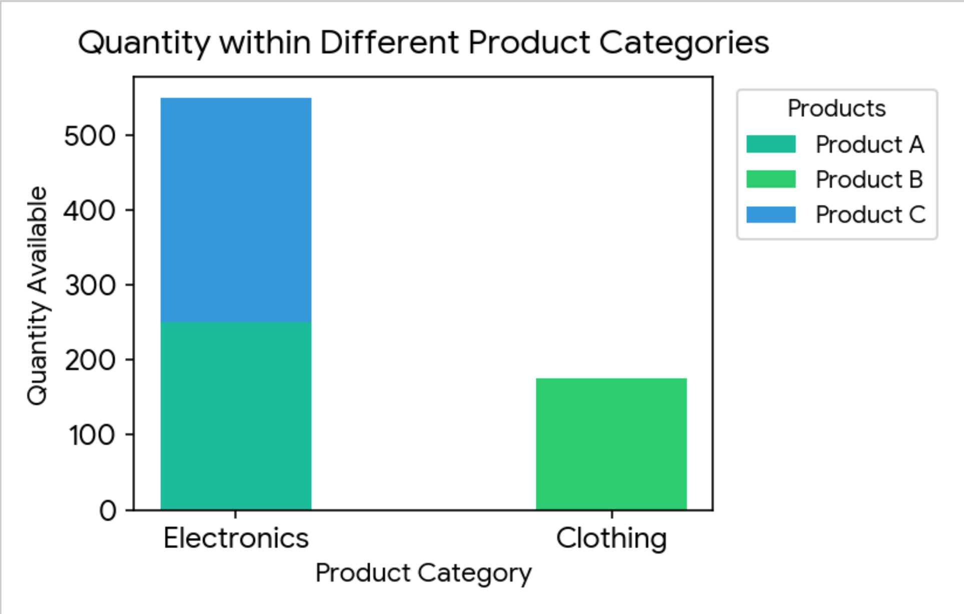
2. Generate a stacked bar chart to show the quantity of each product within different product categories.

R Code:

```
ggplot(inv_data, aes(x=Category, y=Quantity_Available, fill=Product_Name)) +
geom_bar(stat='identity', position='stack', width=0.4) + theme_minimal() +
```

```
labs(title='Quantity within Different Product Categories', x='Product Category', y='Quantity Available')
```

Output / Chart:



3. Build a table to show the inventory data for each product. Label the table elements.

R Code:

```
print(inv_data)
```

Console Screen Output View:

Product Inventory Data Matrix

Product Name	Product ID	Quantity Available	Price ($)
Product A	1	250	$20
Product B	2	175	$15
Product C	3	300	$18

4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Tableau Steps:

```
1. Load 'set18_product_inventory.csv' into Tableau Desktop.
2. Sheet 1: Product Name in Columns, Quantity Available in Rows (Bar Chart).
3. Sheet 2: Category in Columns, Quantity Available in Rows, Color by Product
Name (Stacked Bar Chart).
4. Sheet 3: Product Name, Product ID, Quantity Available, Price in Rows
(Inventory Table).
5. Combine all three sheets into an interactive Dashboard and add Category
filter pickers.
```

Dashboard Mock-up: